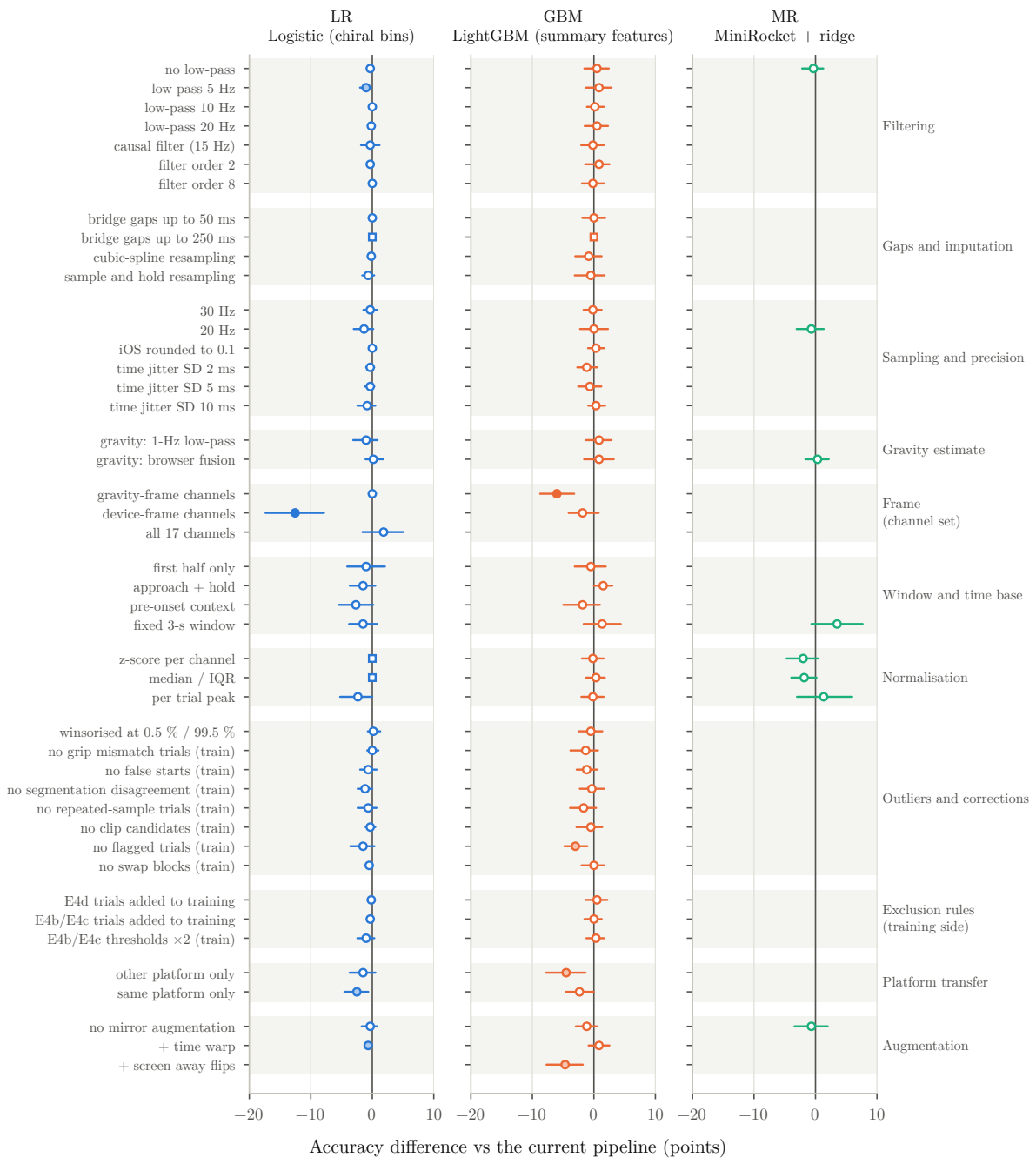
LR
Logistic (chiral bins)
GBM
LightGBM (summary features)
MR
MiniRocket + ridge
no low-pass
low-pass 5 Hz
low-pass 10 Hz
low-pass 20 Hz
causal filter (15 Hz)
filter order 2
filter order 8
Filtering
bridge gaps up to 50 ms
bridge gaps up to 250 ms
cubic-spline resampling
sample-and-hold resampling
Gaps and imputation
30 Hz
20 Hz
iOS rounded to 0.1
time jitter SD 2 ms
time jitter SD 5 ms
time jitter SD 10 ms
Sampling and precision
gravity: 1-Hz low-pass
gravity: browser fusion
Gravity estimate
gravity-frame channels
device-frame channels
all 17 channels
Frame
(channel set)
first half only
approach + hold
pre-onset context
fixed 3-s window
Window and time base
z-score per channel
median / IQR
per-trial peak
Normalisation
winsorised at 0.5 % / 99.5 %
no grip-mismatch trials (train)
no false starts (train)
no segmentation disagreement (train)
no repeated-sample trials (train)
no clip candidates (train)
no flagged trials (train)
no swap blocks (train)
Outliers and corrections
E4d trials added to training
E4b/E4c trials added to training
E4b/E4c thresholds ×2 (train)
Exclusion rules
(training side)
other platform only
same platform only
Platform transfer
no mirror augmentation
+ time warp
+ screen-away flips
Augmentation
−20
−10
0
10
Accuracy difference vs the current pipeline (points)